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Effect of seawater salinity, pH, and temperature on external corrosion behavior and microhardness of offshore oil and gas pipeline: RSM modelling and optimization

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This research aims to investigate the effects of seawater parameters like salinity, pH, and temperature on the external corrosion behaviour and microhardness of offshore oil and gas carbon steel pipes. The immersion tests were performed for 28 days following ASTM G-1 standards, simulating controlled artificial marine environments with varying pH levels, salinities, and temperatures. Besides, Field emission scanning electron microscopy (FESEM) analysis is performed to study the corrosion morphology. Additionally, a Vickers microhardness tester was used for microhardness analysis. The results revealed that an increase in salinity from 33.18 to 61.10 ppt can reduce the corrosion rate by 28%. In contrast, variations in seawater pH have a significant effect on corrosion rate, with a pH decrease from 8.50 to 7 causing a 42.54% increase in corrosion rate. However, the temperature of seawater was found to be the most prominent parameter, resulting in a 76.13% increase in corrosion rate and a 10.99% reduction in the microhardness of offshore pipelines. Moreover, the response surface methodology (RSM) modelling is used to determine the optimal seawater parameters for carbon steel pipes. Furthermore, the desirability factor for these parameters was 0.999, and the experimental validation displays a good agreement with predicted model values, with around 4.65% error for corrosion rate and 1.36% error for microhardness.

Keywords Offshore pipeline, Carbon steel pipe, Pipeline external corrosion, Marine water

Global oil and gas sectors depend on a vast network of carbon steel pipelines for producing and transporting hydrocarbons due to their distinctive properties like high strength, good ductility (yielding behaviour), low weight, cost effectiveness, weldability, and wear resistance[1]. The estimated length of those networks in 2017 was approximately 3.5 million kilometres, which is greater than nine times the distance between the Earth and the moon[2], making them the world's prime logistics infrastructure. Meanwhile, these megaprojects could not avoid failure problems, which were outnumbered by a variety of factors and causes. Among these, corrosion ranks as the second most common cause of metallic pipe failure[3]. Corrosion is the primary cause of metallic pipeline deterioration, damage, and failure, and it may also contribute to economic issues and negatively impact both humans and the environment[4]. Corrosion weakens the mechanical properties of pipes (hardness, toughness, ductility, and so on), increasing the probability of failure due to mechanical breakdowns[5,6]. Corrosion is also considered a crucial factor that influences the maintenance costs and lifespan of pipeline[7].

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Corrosion comes in a variety of forms[8,9] and it is an electrochemical reaction between carbon steel and its internal and external environment[10]. Corrosion can cause internal and external metal loss in pipelines due to a variety of factors, including the properties of pipeline materials, environmental conditions, and the characteristics of the medium that flows inside the pipeline[11]. The characteristics of the medium, such as velocity, temperature, and pressure, commonly cause internal corrosion in pipelines[12]. On the other hand, the environment to which the pipeline is exposed influences external corrosion[13,14].

Offshore marine pipelines that transport hydrocarbons in harsh sea environments are more prone to external corrosion due to the harsh conditions of seawater. Furthermore, there are many factors that contribute to the external corrosion of offshore pipelines, such as seawater microbiological influence, water alkalinity, seawater salinity, and seawater temperature[15,16]. The fluctuation in these sea parameters can substantially reduce the corrosion rate and enhance the pipeline's performance. Therefore, numerous scholarly studies have previously examined the impact of these seawater parameters on corrosion behaviour and pipeline material performance[17–19]. For instance, Wang Xinhua[20] studied the corrosion behaviour of 2Cr13 stainless steel in different artificial seawater environments in 2020, considering pH, temperature, and dissolved oxygen contents. His study revealed that the temperature of seawater had a major impact on the corrosion rate, and a small 10 °C decrease in seawater temperature had an exponential impact on the corrosion rate. Furthermore, in 2020, Smith et al.[21] conducted an immersion test to explore the impact of seawater salinity on mild steel with various chemical compositions. Their study demonstrated that increasing sodium chloride (NaCl) concentrations from 0.05 to 3.5% leads to an increase in corrosion rate, while increasing sodium chloride concentrations to 10% leads to a decrease in corrosion rate. Likewise, in 2019, Darmawan et al.[22] investigated the influence of salinity on aluminium alloys in artificial seawater and reported a similar correlation. The reason for these finding could be a reduction in oxygen solubility accompanying increased water salinity, subsequently lowering water conductivity and, consequently, diminishing the corrosion rate[23,24]. Moreover, Chen et al.[25] examined the corrosion behaviour and mechanical properties of low-alloy steel under different environmental conditions. Their study revealed a substantial decline in the mechanical properties (Ultimate tensile strength (UTS), Yield strength (YS), and elongation to failure (EFL)) of low-alloy steel due to corrosion in varying environments. Gao et al.[26] investigated how saline water cathodic protection potentials altered the microhardness of Q235 steel using electrochemical impedance spectroscopy on steel samples. Their observations also indicated a decrease in microhardness over time. Despite the number of scholarly works presented in the literature about the effect of seawater parameters on corrosion inhibition of offshore pipelines, the combined impact of these parameters has not been well defined so far. Therefore, this research aims to fill that gap by studying the combined effect of these seawater parameters (salinity, pH, and temperature) on the corrosion behaviour and microhardness of oil and gas pipelines. Additionally, RSM modelling is applied to find out the optimal values of these parameters at which the oil and gas pipeline material has optimal performance.

Materials and methods

The methodological framework used to investigate the effects of seawater parameters such as salinity, pH, and temperature on corrosion behaviour and microhardness can be seen in the Fig. 1. However, about materials and the concise techniques are provided in subparts of this section.

Materials

ASTM 106 grade B carbon steel pipe having an inside diameter of 76.2 mm and an outside diameter of 88.9 mm is used in this study. The chemical composition and mechanical properties of pipeline are shown in Tables 1 and 2.

Sample preparations

In this research, a pipe made of carbon steel with a wall thickness of 12.7 mm and a diameter of 3 inches is utilized. Using an abrasive wet-cutting machine, 1-foot-long samples of identical length were cut from each specimen. The external coating was then removed, capped with plastic caps, and sealed on both sides with a chemically resistant material to prevent leakage to the inner layer of the pipes and to guarantee that only the external surface area of the specimen is exposed to the artificial marine environment (see Fig. 2). Moreover, smaller coupon samples, measuring approximately 10 mm in length and 7.5 mm in width, were made for further analysis via FESEM characterization and microhardness evaluation. To prepare these coupons, a series of sandpapers ranging from 120 to 2400 grits were systematically employed using grinding and polishing machinery, ensuring a smooth surface.

Immersion test

In this study, a series of artificial seawater solutions with varying pH, salinity, and temperature were developed. Table 3 provides the ranges of the above-mentioned parameters and experimental conditions. These artificial environments replicate the conditions of the Malaka Strait and the South China Sea[27,28]. Furthermore these samples were immersed for 28 days following ASTM G-1 standards[29]. The temperature was controlled by electric temperature controller (see Fig. 3) and ranges were also selected for distinct seasons under identical marine conditions. In accordance with ASTM TM0169/G31 standards[30], each test involved the exposure of two identical test samples along with coupons for FESEM and microhardness study. The immersion box was left open. Whereas by frequently adding the proper solution, the evaporation losses were maintained between + 1% and − 1% of the initial volume.

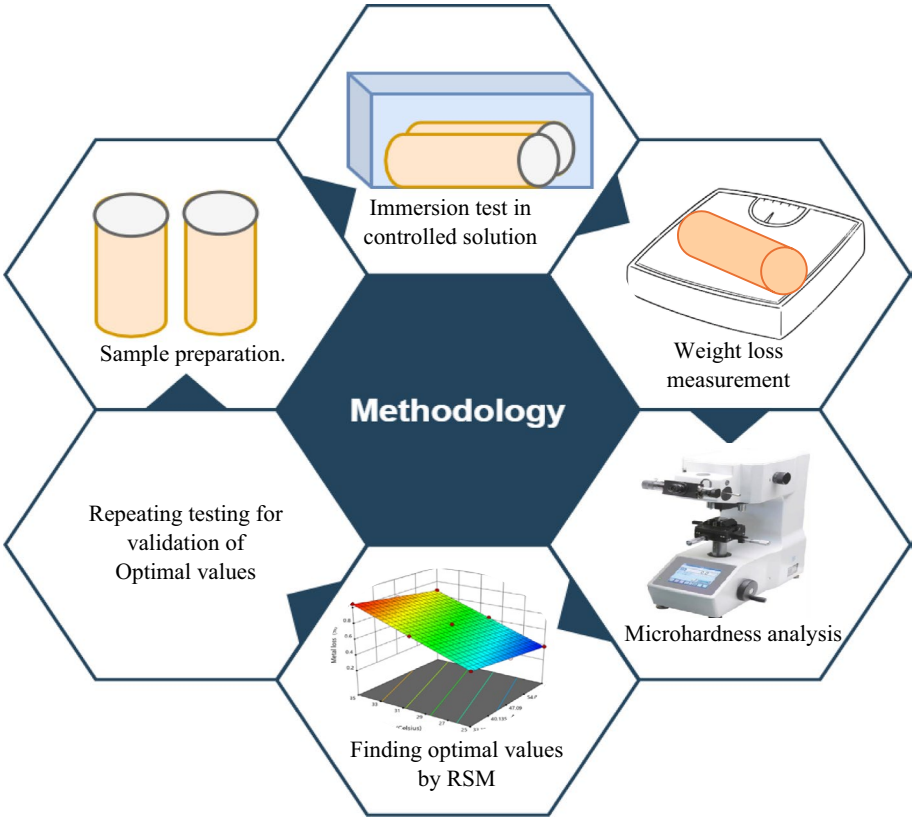


Figure 1. Methodological frame work.

Element	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Fe
Quantity (%)	0.30	1.06	0.035	0.035	0.10	0.40	0.40	0.40	0.15	97.12

Table 1. The composition of carbon steel grade B pipe.

Tensile strength (MPa)	Microhardness (HV)	Elongation (%)	Yield strength (MPa)
455	148	21	240

Table 2. Mechanical properties of ASTM 106 grade B carbon steel pipe.

Weight loss measurement and corrosion rate

After 28 days, the samples were taken out from the solutions and the rust was removed in accordance with ASTM G-31 standards[31], so that only the corroded portion of the samples could be removed without affecting the pipe surface. The weight loss study of corrosion rate was conducted by comparing the pre- and post-exposure sample weights. Following the below equation, the rate of corrosion can be determined[31].

$$Corrosion\ rate = \frac{(K \times W)}{A \times T \times D} \quad (1)$$

where A = pipe exposure area, T = immersion duration in hours, K is constant = 87,600, D = density of sample that is 7.85 g/cm^3 and W = weight loss in grams.

Microhardness measurement

After a 28-day exposure to various artificial marine environments, the microhardness of all samples was assessed following the standardized test method (ASTM E92-82) using a Vickers hardness tester[32]. This assessment utilized a 300-gf load with a 25-s dwell time. Throughout the hardness test conducted at room temperature (25 °C), measurements were taken four times at different locations on each test sample, and the mean values derived from these measurements were used to determine the microhardness. Notably, the experimental error during the hardness assessment remained below 5%.